Stanisław Szufa

Teaching Statement

Teaching Philosophy

In my research, I prove properties of voting rules, and I also run them on real and synthetic data. I would like my students to do both.

- *Concepts come with data.* When I introduce a definition, students will also apply it to a concrete instance. For example, students can load a real participatory budget from Pabulib, compute the outcomes of several voting rules, and only then study the axioms that explain why the outcomes differ.
- *Open-ended problems.* Alongside standard exercises, I will set problems without a known answer, such as finding the statistical model of votes that best describes a city's ballots. Students then have to choose a method themselves and argue for their conclusions.
- *Experiments in the age of language models.* Language models now write much of the code for experiments, so the hard part has moved to designing an experiment and checking that the code really runs it. I will teach students to do both, for example by asking them to find the error in a plausible-looking experiment generated by a language model.

Teaching and Mentoring Experience

At Jagiellonian University (2018–2021), I taught *Introduction to Programming* (a full-year course), *Logic and Set Theory*, and *Linear Algebra* (one semester each) to groups of about twenty undergraduate students. Groups with very different preparation taught me to explain one idea in several ways and to notice early who is falling behind. For my research community, I co-authored the *Guide to Numerical Experiments on Elections* [C15], developed `mapof`, co-developed `prefsampling`, and presented the tutorial *Experiments in Social Choice* at IJCAI 2023, AAMAS 2023, and AAAI 2024. In 2026, I gave a hands-on workshop for newcomers to social choice at the University of Chicago's Data Science Institute. I have also given more than thirty research talks, which has made me comfortable explaining technical material to mixed audiences.

I have co-supervised more than ten BSc and MSc students, and several of these projects led to papers at AAAI 2025 [C19] and ECAI 2025 [C22]. I start each student with a well-defined question for which data and tools are at hand, so that a first result comes quickly, and then move to an open problem whose answer I do not know. As President of the PhD Students' Association at Jagiellonian University, I led an elected body representing doctoral students, which showed me how much a student's success depends on conditions outside the classroom.

Teaching at EPFL

I am prepared to teach core bachelor courses (introductory programming, discrete mathematics, algorithms) and master courses in artificial intelligence and algorithmic game theory. In large bachelor courses, I would use weekly programming and problem-solving exercises with quick feedback, automated where possible, and short problems during lectures that show me which ideas students have not yet understood. Exams would test both proofs and computation. I would like to develop two new courses that complement *Algorithmic Game Theory* (MGT-435) and *Technologies for Democratic Society* (CS-234):

- *Algorithms for Collective Decisions* (master): voting, committee elections, participatory budgeting, and fair division, with real data throughout and a final project on a real participatory budget, possibly with a Swiss municipality.
- *AI and Democracy* (master seminar): AI-mediated deliberation, AI-written summaries of public consultations, and AI agents that vote on people's behalf, open to students from computer science, data science, and digital humanities.

I would also draw on EPFL's Teaching Support Center and mid-semester student feedback.